LUKMAN HAKIM (he/him)

Nature-based Solutions Geospatial Professional

M.Sc. Photogrammetry and Geoinformatics | DAAD Scholarship Awardee

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PROFILE

Geospatial professional with 6+ years of experience delivering GIS and remote-sensing analyses for conservation, forest carbon, blue carbon, ecosystem monitoring, and public-sector projects. Experienced in processing satellite imagery and heterogeneous geospatial datasets using Python, Rasterio, GeoPandas, Google Earth Engine, QGIS, ArcGIS Pro, and PostGIS. Strong focus on reproducible workflows, spatial data quality assurance, technical documentation, and translating scientific methodologies into reliable geospatial outputs. Comfortable working autonomously and collaborating remotely with scientists, government partners, and interdisciplinary teams.

CORE COMPETENCIES

Remote Sensing & Spatial Modelling

Aerial Imagery · Satellite Imagery · LiDAR · Google Earth Engine · ENVI · SNAP · Image classification & quantification · Machine learning

Data Collection and Management

GNSS · Total Stations · Field Survey · UAS · Point Cloud · SQL · PostgreSQL/PostGIS · Relational Model · Metadata documentation

Programming & Automation

Python · TensorFlow · Rasterio · Pandas · PyTorch · API-based data processing · Automation · Reproducible workflows

GIS & Spatial Analysis

QGIS · ArcGIS Pro · Trimble · Cartography · CRS · Route Mapping Conservation planning · Zoning support · Environmental monitoring

WebGIS & Geospatial Services

GeoServer · GeoDjango · Leaflet · Flask · JavaScript · JSON · REST APIs · OGC services

Tools & Collaboration

Git · Docker · FME · Technical documentation · Interdisciplinary teamwork · Stakeholder communication · Capacity Building

PROFESSIONAL EXPERIENCE

Landesamt für Geoinformation und Landentwicklung Baden-Württemberg — Student Job **(06/2025 – 11/2025)**

Identification of green roof areas in Baden-Württemberg using LoD2 building data and PlanetScope imagery

- Batch processed LoD2 building geometry (slope, aspect, area, height), semantic building attributes, and PlanetScope satellite imagery to generate a 2025 roof-plane-level green roof inventory for Baden-Württemberg.
- Developed Python workflows for geodata preparation, feature extraction, quality control, and machine-learning-ready spatial datasets.
- Contributed to climate- and green-infrastructure-relevant mapping by supporting the identification of vegetated roof areas from high-resolution geospatial and satellite data.

The Nature Conservancy Indonesia — GIS & Remote Sensing Specialist **(07/2019 – 08/2024)**

- Supported cross-NGO and government collaborations in designing Marine Protected Area networks using MARXAN, gap analysis, and spatial prioritization to inform local, national, and transboundary zoning aligned with Aichi biodiversity conservation targets.
- Developed GIS and remote-sensing workflows for mangrove mapping, including distribution, canopy density, and aboveground biomass estimation to support forest carbon inventory, social forestry initiatives, and REDD+ readiness in Indonesia.
- Developed remote-sensing workflows for benthic habitat classification, including coral, seagrass, algae, sand, rubble, and rock classes, supporting blue carbon initiative
- Applied multi-temporal satellite datasets, including Maxar, SPOT, Landsat, Sentinel, PlanetScope, and Planet NICFI, using Google Earth Engine for environmental monitoring, land-cover analysis, and spatial modelling.
- Planned and conducted aerial data acquisition using small UAS drones and fixed-wing platforms, including flight planning, image triangulation, orthophoto correction, image stitching, and land-use/land-cover classification for coastal community landscapes.
- Integrated heterogeneous geospatial datasets from satellite imagery, field surveys, participatory mapping, administrative boundaries, and open-source geospatial products into structured spatial databases for analysis and decision-making.
- Mentored junior staff in GIS, remote sensing, data management, spatial analysis, and reproducible mapping workflows, improving internal technical capacity across conservation projects.

Research Institute for Tuna Fisheries Bali — Internship **(08/2018 – 09/2018)**

- Collected fisheries-related spatial and environmental data for bachelor thesis research.
- Applied remote sensing and Generalized Additive Model (GAM) methods for distribution mapping of Yellowfin Tuna (*Thunnus albacares)* in the Indian Ocean.

EDUCATION

Master of Science (M.Sc.), Photogrammetry and Geoinformatics — Hochschule für Technik Stuttgart

(2024 – 2026, DAAD Scholarship Awardee)

Bachelor of Science (B.Sc.), Cartography and Remote Sensing — Universitas Gadjah Mada

(2014 – 2019, BIDIKMISI Scholarship Awardee)

LANGUAGES

Javanese: Native | Indonesian: Native | English: C1 | German: B1-B2 | Japanese: Novice

SELECTED PUBLICATIONS

- Aung Kyaw, H., Ayala Rico, J., Awontemi, I. A., **et al**. (2026). Urban Vegetation Mapping from High-Resolution Aerial Imagery with Deep Learning. DGPF Jahrestagung 2026.
- Fajariyanto, Y., Green, A., **Hakim, L.**, et al. (2024). A Resilient Marine Protected Area Network Design: A First for the Arafura–Timor Seas. Coastal Management, 52(3), 147–173.
- Prakoso, D. A., **Hakim, L.**, et al. (2023). The Dynamics of Mangrove and Pond Changes in East Kalimantan, Indonesia. IOP Conference Series: Earth and Environmental Science.
- Fajariyanto, Y., Green, A. L., Ramadyan, F., Suardana, N., **Hakim, L.**, et al. (2019). Designing a Network of Marine Protected Areas for Fisheries Management Area 715 and Six Associated Provinces in Indonesia. Technical report prepared for the USAID Sustainable Ecosystems Advanced Project.
- **Hakim, L.**, Lazuardi, W., et al. (2018). Assessing WorldView-2 Satellite Imagery Accuracy for Bathymetry Mapping in Pahawang Island, Indonesia. IOP Conference Series: Earth and Environmental Science.

ADDITIONAL QUALIFICATIONS

Small UAS Remote Pilot · Open Water Diver · Field-based coastal and marine survey experience

EXTRACURRICULAR

2023	Presenter at 5th Asia Pacific Coral Reef Symposium, NUS Singapore
2023	Participant at Mangrove Monitoring International Training (MMIT), Archipelagic Island State Forum Bali Indonesia
2017	Presenter at World Blue Carbon Conference, Jakarta Convention Center Indonesia
2017	Presenter at Seminar Nasional Geomatika, Geospatial Information Agency Indonesia

Additional early experience includes serving as Vice President of the Geography Study Club at Universitas Gadjah Mada, and winning national college level debate competitions.

REFERENCES

- Prof. Dr. rer. nat. Michael Mommert | Hochschule für Technik Stuttgart | michael.mommert@hft-stuttgart.de
- Maryse Hilemann | Landesamt für Geoinformation und Landentwicklung Baden-Württemberg | Maryse.Hillemann@lgl.bwl.de
- Yusuf Fajariyanto | The Nature Conservancy Indonesia | yfajariyanto@tnc.org